Scalability & Future Plan

SmartBridge / Skill Wallet Project Assessment

DATE	TEAM ID	PROJECT NAME	MAXIMUM MARKS
15 March 2026	6ab2a81f009dcca3125ab2e5	WALL-T	1 Mark

Scalability & Future Plan Overview

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.NO	LIMITATION	IMPACT	PRIORITY TO ADDRESS (HIGH / MEDIUM / LOW)
1	**Single-server / local database architecture**	Limits the system from processing high-volume concurrent user requests simultaneously.	**High**
2	**Basic local caching & storage capacity**	Temporary storage constraints during large-scale historical data queries.	**Medium**
3	**Manual configuration for system deployment**	Slower deployment cycle when deploying to multiple cloud regions or target clients.	**Medium**

Scalability Plan

S.NO	SCALABILITY ASPECT	CURRENT STATE	PROPOSED UPGRADE / SOLUTION
1	**User Load**	Supports small to moderate concurrent active users.	Implement cloud auto-scaling with Load Balancers (AWS Elastic Load Balancing / NGINX).
2	**Data Storage**	Relies on centralized relational database storage.	Migrate to distributed cloud databases (AWS DynamoDB / MongoDB Cluster) with sharding.
3	**Performance**	Direct database queries for every operational request.	Integrate Redis memory caching and CDN integration to minimize response latency.
4	**Security**	Standard role-based access control and static API tokens.	Implement OAuth2.0 authentication, TLS 1.3 encryption, and periodic automated security audits.

Future Roadmap

PHASE	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	EXPECTED IMPACT
Phase 2	Cloud Infrastructure & Dynamic Scaling	3 Months	Ensures system reliability and continuous uptime under heavy user demand.

PHASE	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	EXPECTED IMPACT
Phase 3	Mobile Application & Push Notifications	6 Months	Enhances user accessibility and engagement across iOS and Android devices.
Phase 4	AI Analytics & Automated Optimization	9-12 Months	Provides predictive insights and intelligent automated workflows for end-users.